

ScholarMind - Data Sharing Module

UI/UX Design Handoff

Source of Truth for Presentation Layer

This document defines the UI/UX flow for the Data Sharing module.

The backend architecture has already been implemented and should **NOT** be redesigned. The presentation layer should integrate with the existing backend.

The target audience for this document is:

- UI/UX Design
 - Flutter Frontend Development
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Overall Design Philosophy

The backend has already been architected around a generic export/import pipeline capable of supporting all ScholarMind modules.

The UI should therefore expose this capability in a way that feels simple and intuitive rather than technical.

The primary goals are:

- Minimal user friction
 - Familiar interaction patterns
 - Recognition over recall
 - Reusable architecture for future modules
 - Consistency with the rest of ScholarMind
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Supported Export Targets

The system should support exporting resources from every ScholarMind module.

Initially this includes:

- Notes
- Flashcards
- Countdown Manager

- Grade Tracker (future)
- Study Planner (future)
- Any future module implementing the Data Sharing interface

The UI must therefore be designed to scale naturally as more modules are added.

Supported Export Methods

Three export methods are supported.

1. ScholarMind Archive

`.scholar`

Native ScholarMind archive format.

Primary method for long-term storage and sharing.

2. QR Code

Generate a QR code representing the exported package.

Allows nearby users to scan and import.

3. Share Link

Generate a ScholarMind share link.

Selecting this option should eventually launch the native share sheet after the link has been generated.

Import Philosophy

Import intentionally does **NOT** have a dedicated screen.

Import should feel automatic.

Examples:

- Opening a .scholar file
- Opening a ScholarMind share link
- Scanning a QR code

These actions immediately launch the import workflow.

This avoids unnecessary navigation and follows platform conventions.

Navigation Flow

Home

↓

Share Files

↓

Export Library

↓

Export Cart

↓

Choose Export Method

↓

(.scholar / QR / Link)

↓

Export

Import flow

Receive

↓

Import Preview Dialog

↓

Background Import

↓

Finished

Presentation Layer

The module should consist of the following screens.

Screen 1

Export Library

Purpose

Allow users to browse and select resources for export.

This becomes the primary export screen.

Layout

Resources should be grouped by module.

Example

- ▼ Notes
- ▼ Flashcards
- ▼ Countdown Manager
- ▼ Grade Tracker
- ▼ Study Planner

Each module is collapsible.

Collapsed by default.

This reduces visual clutter while still allowing one unified export page.

Resource Presentation

Resources should **NOT** simply display filenames.

Instead, each resource should contain enough contextual information that the user can immediately recognize it.

This follows the HCI principle:

Recognition rather than Recall

Users should recognize the desired resource rather than attempting to remember filenames.

Example

Instead of

Networking.pdf

Display

Networking Lecture Notes

PDF

Updated Yesterday

12 Pages

Instead of

Folder A

Display

Networking

8 Notes

3 Subfolders

Last Updated Today

Instead of

Deck 1

Display

Networking Flashcards

45 Cards

AI Generated

The exact metadata depends on the module.

Selection

Resources support multi-selection.

Entire modules do not become selected.

Only individual resources.

Folder Behaviour

Selecting a folder should **NOT** immediately export every child.

Instead ScholarMind asks

Include child resources?

Options

Include None

Choose Resources

Include Everything

Choosing resources opens a lightweight selection dialog showing only the folder contents.

This provides maximum flexibility.

Example use case:

Export entire folder except one note.

Search

The Export Library should support searching.

Existing search dialog architecture should be reused where possible.

Continue Button

Once one or more resources have been selected

Continue

navigates to the Export Cart.

Screen 2

Export Cart

Purpose

Allow users to review everything being exported before generating the archive.

Conceptually similar to an e-commerce shopping cart.

Layout

Resources remain grouped by module.

Example

Notes

Networking Folder

Database Notes

Flashcards

Networking Deck

Countdowns

Final Presentation

The hierarchy should remain visible.

Metadata

Metadata already produced by the backend should be displayed.

Examples

Networking Folder

8 Notes

3 Subfolders

Networking Deck

45 Cards

This improves confidence before exporting.

Editing

Users should still be able to modify selections.

Support:

Remove

Swipe to remove

Delete button

Returning to the previous page should not be required for small edits.

Continue

Proceed to export method selection.

Bottom Sheet

Choose Export Method

A bottom sheet is preferred over another screen.

Methods

`.scholar`

QR Code

Share Link

Each method begins its own workflow.

Scholar Archive Flow

Selecting

`.scholar`

Immediately generates the archive.

Once complete

Launch native file save dialog.

No additional ScholarMind screen is required.

QR Flow

If QR generation is fast

Immediately display the QR screen.

If generation requires noticeable time

Display a loading screen.

Do not use the AppBar for QR generation because the user is actively waiting for the QR code.

Screen 3

QR Display

Displays

Generated QR

Actions

Download Image

Share Image

Close

Regeneration is not required.

The QR should remain valid until the export session ends.

Share Link Flow

Generate the share link.

When complete

Launch the native operating system share sheet.

The user can then

Copy

Message

Email

etc.

Import Workflow

Import begins automatically whenever ScholarMind receives

- .scholar file
 - ScholarMind share link
 - QR scan
-

Dialog

Import Preview

Purpose

Allow users to verify imported resources before committing.

Example

Networking Package

Folders

2

Notes

15

Flashcards

45

Countdowns

3

Buttons

Import

Cancel

Background Import

Import runs in the background.

The AppTaskBar should be used because importing large packages may require significant processing.

Conflict Resolution

If imported resources already exist

Display a conflict dialog.

Supported actions

Rename

Replace

Keep Both

Skip

Rename should allow immediate editing before continuing.

Export History (Future Enhancement)

Not required for Version 1 but the UI should be designed so this can be added later.

Purpose

Reuse previously generated exports.

Example

Networking Study Pack

Yesterday

Share Again

This supports sharing the same package with multiple friends without rebuilding the export.

Home Screen Integration

A new button should be added to the Home screen.

Share Files

Selecting this button opens the Export Library.

Module-Level Sharing

Future enhancement.

Individual modules may eventually expose

Share

from overflow menus.

This is intentionally deferred to reduce implementation effort.

The current implementation should only support Home screen entry.

HCI Principles Applied

The presentation layer should follow these principles throughout.

Recognition Rather Than Recall

Provide contextual metadata rather than filenames alone.

Users should recognize resources immediately.

Progressive Disclosure

Keep module groups collapsed by default.

Reveal details only when requested.

This keeps the page manageable even as ScholarMind grows.

User Control and Freedom

Allow users to:

- Remove selections
- Modify selections
- Choose folder contents
- Resolve conflicts

Avoid forcing users to restart workflows.

Visibility of System Status

Display loading indicators whenever archive generation or QR generation requires noticeable time.

Use the AppTaskBar for long-running background imports.

Consistency

The export flow should mirror the interaction patterns already established in ScholarMind's Notes and Quiz modules wherever practical.

Expected Presentation Files

The current UI is expected to require approximately the following presentation components.

`export_library_screen.dart`

`export_cart_screen.dart`

`qr_display_screen.dart`

Dialogs

`folder_selection_dialog.dart`

`import_preview_dialog.dart`

`conflict_resolution_dialog.dart`

Bottom Sheet

`export_method_bottom_sheet.dart`

Loading

`export_loading_dialog.dart`

The native operating system should handle:

- Save File
- Share Sheet

These do not require additional ScholarMind screens.

Development Notes

The backend architecture has already been completed and should be reused.

The frontend should simply orchestrate the existing backend pipeline:

Export Library

↓

Resource Selection

↓

Export Cart

↓

Export Method

↓

Existing ExportService

↓

ArchiveBuilderService

↓

Output

The presentation layer should not duplicate business logic that already exists within the backend.

Final Assessment

This UX specification defines the complete presentation layer for the Data Sharing module while intentionally leaving visual styling to the UI design phase. It should provide enough detail for a UI designer to produce high-fidelity mockups and for Codex to implement the Flutter presentation layer directly against the existing backend architecture with minimal ambiguity. I would consider this the presentation-layer source of truth going forward.